

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

Regular & Supplementary Summer 2025

**Course: B. Tech.      Branch : Electrical Engineering**

**Semester : I**

**Subject Code & Name: 24AF1000ES106B & Basic Electrical and Electronics Engineering**

**Max Marks: 60**

**Date:24/07/25**

**Duration: 3 Hr.**

**Instructions to the Students:**

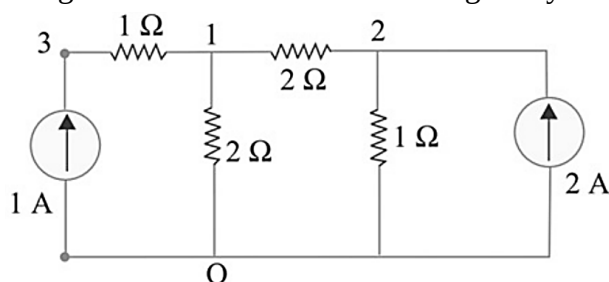
1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

|             |                                                                                                                                                                              | (Level/CO) | Marks     |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|
| <b>Q. 1</b> | <b>Objective type questions. (Compulsory Question)</b>                                                                                                                       |            | <b>12</b> |
| 1           | Which of the following is not a passive circuit element?<br>a) Resistor<br>b) Capacitor<br>c) Inductor<br>d) Transistor                                                      | Analyze    | 1         |
| 2           | Kirchhoff's Current Law is based on the principle of:<br>a) Conservation of charge<br>b) Conservation of energy<br>c) Conservation of momentum<br>d) Conservation of voltage | Understand | 1         |
| 3           | In nodal analysis, which law is used primarily?<br>a) Ohm's Law<br>b) Kirchhoff's Current Law (KCL)<br>c) Kirchhoff's Voltage Law (KVL)<br>d) Faraday's Law                  | Apply      | 1         |
| 4           | Reactive power is measured in:<br>a) Watts<br>b) Volts<br>c) VAR<br>d) Ohms                                                                                                  | Understand | 1         |
| 5           | The main difference between a generator and a motor is:<br>a) Type of winding<br>b) Direction of current<br>c) Type of core<br>d) Supply connection                          | Analyze    | 1         |
| 6           | Which motor works on the principle of rotating magnetic field?<br>a) DC motor<br>b) Stepper motor<br>c) Induction motor<br>d) Servo motor                                    | Understand | 1         |
| 7           | In a single-phase transformer, the core is laminated to:<br>a) Reduce copper loss                                                                                            | Apply      | 1         |

- b) Increase eddy current loss  
c) Reduce eddy current loss  
d) Strengthen the core
- 8 The output frequency of a full-wave rectifier is: Apply 1  
a) Same as input AC  
b) Half of input AC  
c) Double of input AC  
d) Zero
- 9 In a common-emitter configuration, the output is taken across: Apply 1  
a) Emitter and Base  
b) Collector and Emitter  
c) Base and Collector  
d) Emitter and Ground
- 10 Which of the following is not a functional element of an instrument? Analyze 1  
a) Transducer  
b) Signal conditioning  
c) Power supply  
d) Data presentation
- 11 Which of the following instruments works on the principle of the interaction of magnetic fields? Understand 1  
a) Moving Coil  
b) Moving Iron  
c) Digital Voltmeter  
d) Energy Meter
- 12 Which instrument can measure both AC and DC currents? Understand 1  
a) PMMC  
b) Moving Iron  
c) Digital Ammeter  
d) Thermocouple

**Q. 2 Solve the following.**

- A) State and explain Kirchhoff's Laws with examples. Understand 12  
B) Determine the voltages 1 and 2 of the network in Figure by nodal analysis. Apply 6



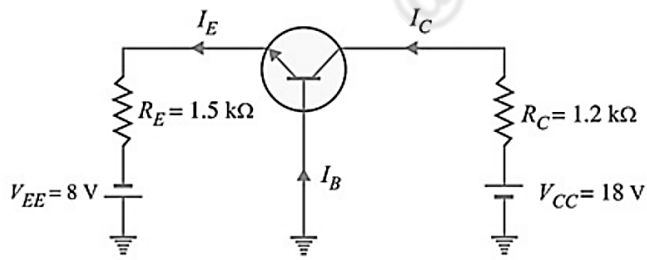
**Q.3 Solve the following.**

- A) Difference between Generator and Motor. Analyze 12  
B) Explain the working principle of Induction motor. Understand 6

**Q. 4 Solve Any Two of the following.**

- A) Draw and explain the V-I (voltage-current) characteristics of a PN junction diode. Understand 12  
B) Explain the working of a center tapped full-wave rectifier with a neat diagram. Understand 6  
C) Explain the working of Zener diode. Understand 6

- Q.5 Solve Any Two of the following.** 12
- A)** For the common base circuit shown in Figure, determine  $I_C$  and  $V_{CB}$ . Assume the transistor to be of silicon. Apply 6



- B)** Describe the working of a NPN transistor with the help of a circuit diagram. Understand 6
- C)** With the help of a circuit diagram, explain the operation of a transistor as a switch. Apply 6
- Q. 6 Solve Any Two of the following.** 12
- A)** Write a short note on a Function Generator. Understand 6
- B)** Explain the working of a Digital Storage Oscilloscope with a neat diagram. Understand 6
- C)** Explain the working of a Multi-meter. Understand 6

\*\*\* End \*\*\*